User Manual: `Parallelized Gaussian Process Interpolative Decomposition`

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1 Introduction

This manual is a user guide to use `Parallelized Gaussian Process Interpolative Decomposition` (`P-GP-ID`) enabling **efficient frequency sweep** for `VWT-IE-Solver-Unified-DG.x` solver. This code pre-requisites `Python` with installed packages `numpy`, `scipy`, and `scikit-learn`. Also, the current version is **alpha-version**, indicating upcomming updates may lead many parts of this manual to be deprecated.

2 Installation

`P-GP-ID` is written in `Python`, and is recommended to be run under virtual environments such as `anaconda`. Once `anaconda` is installed, user may create an environment (in this example named as `myenv`) with required packages and then activate it.

```
conda create -n myenv numpy scipy scikit-learn
conda activate myenv
```

3 Setting up

`P-GP-ID` itself is not a solver, but a Gaussian process-based tool that enables advanced frequency-interpolation with active sampling. Currently, `VWT-IE-Solver-Unified-DG.x` is embedded as the backend sover to provide the high fidelity solutions. To set up the problem of interest, user needs to create setup a simulation directory according to `VWT-IE-Solver-Unified-DG.x` user manual. This directory then needs to go under `data/VWTdata` directory.

This created directory will be supplied as an argument to the `main.py`. Otherwise, it will automatically fall back to its default setting: `./data/VWT-data/circylinder`

4 Running

To run the program using default settings:

```
python main.py
```

To see all available arguments:

```
python main.py -h
```

Descriptions on each argument are in the following section.
Following is the list of default arguments:

	Value	Meaning
`-p`	./data/VWT-data/circylinder	simulation directory
`-m`		model name (`*.domain`) is automatically found
`-A`	0 180 181	angular sweep settings: `np.linspace(0, 180, 181)`
`-S`	0	phi-sweep
`-F`	100 300	frequency band will be set to [100, 300] MHz
`-v`		no validation
`-a`	0	acquisition function: maximum variance
`-x`	1	x-data normalization strategy: z-score (standard normalization)
`-n`	3	number of initial frequency samples
`--add`	1	added frequency samples per iteration
`-t`	3	number of terms used for Gaussian process kernel
`-s`	1	sampling strategy: Latin Hypercube Sampling
`-i`	20	maximum iteration for Gaussian Process
`-T`	1e-5	IVR tolerance; tolerance for Gaussian Process

5 Arguments

This section dedicates to explain each argument given to `P-GP-ID`.

- `-p`, `--path-simulation`:
  Used to specify the simulation directory.
- `-m`, `--model-name`:
  It is the name of the simulation under the simulation directory, which is the name of `.domain` file. For instance, if user has set up a simulation under `sim` directory with `model.domain` under the directory, then the correct argument is "model." If not provided, the code will attept to find `.domain` file under the simulation directory. If there is more than one `.domain` file, the automatic search fails with a `Runtime Error`.
- `-A`, `--angle`, `--angle-settings`:
  Requires 3 arguments: [start, end, number] of angles to sweep. For example, [0, 100, 5] will create [0, 25, 50, 75, 100]
- `-S`, `--ast`, `--angular-sweep-type`:
  Type of angular sweep. Currently, the code relies on a single angular sweep either ϕ- or θ-sweep. 0: ϕ-sweep, 1: θ-sweep
- `-F`, `--freq`, `--frequency-settings`:
  Requires 3 arguments: [start, end, number] of frequencies. All frequency samples will be bound to this limit. "number" will be used as number of samples for calculating acquisition function, final output, and validation (if enabled).
- `-v`, `--no-validate`:
  For Debugging Purpose. Flag to validate. At the end of all process, `P-GP-ID` will attempt to validate the accuracy by calculating RMSE of the prediction and ground-truth for grid-sampled inputs. This means, if the corresponding grid-sampled ground-truth is not prepared, the code will attempt to run simulations over the entire validation points.

- `-a`, `--ax-fx`, `--acquisition-function`:
  Type of acquisition function for adaptive sampling. An integer $\in [0, 2]$ can be given. User can choose between [maximum variance, expected improvement, upper confidence bound]. Defaults to 0.
- `-n`, `--n-init`:
  Initial number of frequency sampling. The initial samples are always sampled with Latin Hypercube Sampling. Defaults to 3.
- `--add`, `--freq-add`:
  Number of added frequency samples per Gaussian Process iteration. Defaults to 1.
- `-t`, `--terms`:
  Terms. Currently stacked RBFP kernel is used, where terms will define the number of stacks.
- `-s`, `--sample`, `--sampling-strategy`:
  Type of sampling strategy. 0 or 1 can be given. User can choose between [Grid Sampling, Latin Hypercube Sampling]. Defaults to 0.
- `-i`, `--max-iter`:
  Maximum iteration for Adaptive Sampler. Therefore, the maximum frequency samples will be determined as $\boxed{\texttt{-n}} + \boxed{\texttt{--add}} * \boxed{\texttt{-i}}$.
- `-T`, `--tol`:
  Tolerance for integrated variance reduction (IVR; weighted sum of variances over basis). Defaults to 1e5.